Customer Churn Analysis - SQL Queries with Output

Each section shows the business question, the SQL query, a screenshot of the result, and the insight. Table: customers (7,200 rows, cleaned data). Engine: SQLite.

DQ1 - Data quality: row count, unique ids, invalid values (all problem counts must be 0)

```
SELECT COUNT(*) AS rows_, COUNT(DISTINCT customer_id) AS distinct_customers,
       SUM(CASE WHEN monthly_charges<=0 THEN 1 ELSE 0 END) AS bad_monthly_charges,
       SUM(CASE WHEN total_charges IS NULL OR total_charges=0 AND tenure_months>0 THEN 1 ELSE 0 END) AS
    bad_total_charges,
       SUM(CASE WHEN contract NOT IN ('Month-to-month','One year','Two year') THEN 1 ELSE 0 END) AS bad_contract FROM
    customers;
```

Output (screenshot):

rows_	distinct_customers	bad_monthly_charges	bad_total_charges	bad_contract
7,200	7,200	0	0	0

Insight: All checks pass: 7,200 unique customers, no invalid charges or contract labels - safe to analyse.

Q1 - Executive summary: how big is the churn problem?

```
SELECT COUNT(*) AS customers, SUM(churned) AS churned, ROUND(100.0*SUM(churned)/COUNT(*),1) AS churn_rate_pct,
       ROUND(SUM(monthly_charges),0) AS monthly_revenue, ROUND(SUM(CASE WHEN churned=1 THEN monthly_charges END),0) AS
    churned_monthly_revenue,
       ROUND(100.0*SUM(CASE WHEN churned=1 THEN monthly_charges END)/SUM(monthly_charges),1) AS pct_revenue_churned,
    ROUND(AVG(tenure_months),1) AS avg_tenure_months
FROM v_customers;
```

Output (screenshot):

customers	churned	churn_rate_pct	monthly_revenue	churned_monthly_revenue	pct_revenue_churned	avg_tenure_months
7,200	1,741	24.2	4,977,351	1,360,021	27.3	36.2

Insight: 24.2% of customers churned and they carried 27.3% of monthly revenue - churn is a revenue problem, not just a customer-count problem.

Q2 - Churn by contract type, with each contract's share of ALL churners (window function over an aggregate)

```
SELECT contract, COUNT(*) AS customers, SUM(churned) AS churned, ROUND(100.0*AVG(churned),1) AS churn_rate_pct,
       ROUND(100.0*SUM(churned)/SUM(SUM(churned)) OVER (),1) AS share_of_all_churners_pct
FROM v_customers GROUP BY contract ORDER BY churn_rate_pct DESC;
```

Output (screenshot):

contract	customers	churned	churn_rate_pct	share_of_all_churners_pct
Month-to-month	3,112	1,274	40.9	73.2
One year	1,877	299	15.9	17.2
Two year	2,211	168	7.6	9.6

Insight: Month-to-month customers churn at 40.9% vs 7.6% on two-year plans and make up 73% of all churners.

Q3 - Churn by tenure band: when do customers leave?

```
SELECT tenure_band, COUNT(*) AS customers, ROUND(100.0*AVG(churned),1) AS churn_rate_pct,
    ROUND(AVG(monthly_charges),0) AS avg_monthly_charge
FROM v_customers GROUP BY tenure_band ORDER BY tenure_band;
```

Output (screenshot):

tenure_band	customers	churn_rate_pct	avg_monthly_charge
1. 0-6 m	586	43.9	693
2. 7-12 m	632	40.3	694
3. 13-24 m	1,187	31.8	683
4. 25-48 m	2,418	23.7	697
5. 49+ m	2,377	11.7	688

Insight: Churn falls steadily with tenure: 43.9% in the first 6 months vs 11.7% after 4 years - the first year is the danger zone.

Q4 - Contract x internet service matrix (conditional aggregation = pivot in plain SQL)

```
SELECT contract,
  ROUND(100.0*AVG(CASE WHEN internet_service='Fiber optic' THEN churned END),1) AS fiber_churn_pct,
  ROUND(100.0*AVG(CASE WHEN internet_service='DSL' THEN churned END),1) AS dsl_churn_pct,
  ROUND(100.0*AVG(CASE WHEN internet_service='No internet' THEN churned END),1) AS no_internet_churn_pct,
  SUM(CASE WHEN internet_service='Fiber optic' THEN 1 ELSE 0 END) AS fiber_customers
FROM v_customers GROUP BY contract ORDER BY fiber_churn_pct DESC;
```

Output (screenshot):

contract	fiber_churn_pct	dsl_churn_pct	no_internet_churn_pct	fiber_customers
Month-to-month	56.3	34.4	22.2	1,322
One year	25.2	10.9	5.5	818
Two year	12.2	5.5	1.8	973

Insight: Month-to-month Fiber optic customers churn at 56.3%; moving customers onto longer contracts cuts this sharply (one-year 25.2%, two-year 12.2%).

Q5 - Do add-on services reduce churn? (internet customers only)

```
SELECT n_addons, COUNT(*) AS customers, ROUND(100.0*AVG(churned),1) AS churn_rate_pct, ROUND(AVG(monthly_charges),0)
    AS avg_monthly_charge
FROM v_customers WHERE internet_service <> 'No internet' GROUP BY n_addons ORDER BY n_addons;
```

Output (screenshot):

n_addons	customers	churn_rate_pct	avg_monthly_charge
0	294	30.3	653
1	1,141	29.5	733
2	1,781	29.1	789
3	1,457	25.8	858
4	715	25.7	924
5	169	27.8	994
6	20	15	1,039

Insight: Add-on count has only a weak effect (30% -> ~26%): quantity of services does not protect customers as much as contract type and service quality.

Q6 - Payment method x paperless billing

```
SELECT payment_method, paperless_billing, COUNT(*) AS customers, ROUND(100.0*AVG(churned),1) AS churn_rate_pct
FROM v_customers GROUP BY payment_method, paperless_billing ORDER BY churn_rate_pct DESC LIMIT 6;
```

Output (screenshot):

payment_method	paperless_billing	customers	churn_rate_pct
Electronic check	Yes	1,778	30.1
Electronic check	No	594	26.6
Bank transfer	Yes	876	23.2
Mailed check	Yes	721	23
Credit card	Yes	807	22.4
Mailed check	No	762	21.3

Insight: Electronic-check customers on paperless billing churn the most (30.1%): a sign of payment friction / lower commitment.

Q7 - Support tickets in the last 6 months vs churn (service-quality signal)

```
SELECT CASE WHEN support_tickets_6m=0 THEN '0' WHEN support_tickets_6m=1 THEN '1' WHEN support_tickets_6m=2 THEN '2'
    ELSE '3+' END AS tickets_6m,
       COUNT(*) AS customers, ROUND(100.0*AVG(churned),1) AS churn_rate_pct
FROM v_customers GROUP BY 1 ORDER BY 1;
```

Output (screenshot):

tickets_6m	customers	churn_rate_pct
0	2,634	15.2
1	2,295	23.2
2	1,313	31.4
3+	958	41.3

Insight: Support tickets are a strong early-warning signal: 15.2% churn with no tickets vs 41.3% with 3 or more.

Q8 - Signup-year cohorts: churn rate and average tenure

```
SELECT substr(signup_date,1,4) AS signup_year, COUNT(*) AS customers, ROUND(100.0*AVG(churned),1) AS churn_rate_pct,
    ROUND(AVG(tenure_months),1) AS avg_tenure_months
FROM v_customers GROUP BY 1 ORDER BY 1;
```

Output (screenshot):

signup_year	customers	churn_rate_pct	avg_tenure_months
2020	220	10	70.5
2021	1,231	11.8	63.5
2022	1,277	13.8	51.2
2023	1,216	22	39.1
2024	1,210	29.1	26.9
2025	1,228	35.2	14.7
2026	818	42.4	4.6

Insight: Newer signup cohorts show higher churn, but they are also the youngest customers - cohort comparisons must be read together with tenure (churn risk is highest early).

Q9 - Cities whose churn is above their region's average (CTE + join + RANK window function)

```
WITH city AS (SELECT region, city, COUNT(*) AS customers, AVG(churned) AS rate FROM v_customers GROUP BY region,
    city),
     reg  AS (SELECT region, AVG(churned) AS region_rate FROM v_customers GROUP BY region)
SELECT c.region, c.city, c.customers, ROUND(100*c.rate,1) AS city_churn_pct, ROUND(100*r.region_rate,1) AS
    region_churn_pct,
       ROUND(100*(c.rate-r.region_rate),1) AS gap_pts, RANK() OVER (PARTITION BY c.region ORDER BY c.rate DESC) AS
    rank_in_region
FROM city c JOIN reg r USING(region) WHERE c.rate > r.region_rate ORDER BY gap_pts DESC LIMIT 8;
```

Output (screenshot):

region	city	customers	city_churn_pct	region_churn_pct	gap_pts	rank_in_region
East	Kolkata	321	26.2	23.6	2.6	1
North	Chandigarh	368	28.8	26.3	2.5	1
West	Mumbai	537	26.4	24.5	1.9	1
West	Pune	495	25.5	24.5	0.9	2
South	Hyderabad	568	23.6	22.8	0.8	1
South	Chennai	573	23.2	22.8	0.4	2

Insight: Cities above their regional average are priority areas for local service and network quality checks.

Q10 - Lifetime revenue: churned vs retained customers

```
SELECT churn, COUNT(*) AS customers, ROUND(AVG(tenure_months),1) AS avg_tenure_months, ROUND(AVG(total_charges),0) AS
    avg_lifetime_revenue, ROUND(AVG(monthly_charges),0) AS avg_monthly_charge
FROM v_customers GROUP BY churn;
```

Output (screenshot):

churn	customers	avg_tenure_months	avg_lifetime_revenue	avg_monthly_charge
No	5,459	39.2	25,740	663
Yes	1,741	27	20,958	781

Insight: Churned customers leave with lower lifetime revenue because they leave early (shorter tenure), at similar monthly prices.

Q11 - Which small segments produce most churners? (contract x internet x first-year, segments >= 100 customers)

```
SELECT contract, internet_service, CASE WHEN tenure_months<=12 THEN 'first year' ELSE 'after 1st year' END AS stage,
    COUNT(*) AS customers,
       ROUND(100.0*AVG(churned),1) AS churn_rate_pct, ROUND(100.0*SUM(churned)/(SELECT SUM(churned) FROM
    v_customers),1) AS pct_of_all_churners
FROM v_customers GROUP BY 1,2,3 HAVING COUNT(*) >= 100 ORDER BY pct_of_all_churners DESC LIMIT 6;
```

Output (screenshot):

contract	internet_service	stage	customers	churn_rate_pct	pct_of_all_churners
Month-to-month	Fiber optic	after 1st year	993	51.1	29.1
Month-to-month	DSL	after 1st year	834	29.5	14.1
Month-to-month	Fiber optic	first year	329	72	13.6
One year	Fiber optic	after 1st year	722	23.4	9.7
Month-to-month	DSL	first year	257	50.2	7.4
Month-to-month	No internet	after 1st year	531	19.4	5.9

Insight: A few segments produce most churn: month-to-month Fiber customers after year 1 alone are 29% of all churners; in their first year they churn at 72%.

Q12 - Revenue-weighted priority: top 10 ACTIVE customers on month-to-month Fiber with 3+ tickets (retention call list)

```
SELECT ROW_NUMBER() OVER (ORDER BY monthly_charges DESC) AS priority, customer_id, region, tenure_months,
    monthly_charges, support_tickets_6m, late_payments_12m
FROM v_customers WHERE churned=0 AND contract='Month-to-month' AND internet_service='Fiber optic' AND
    support_tickets_6m>=3 ORDER BY priority LIMIT 10;
```

Output (screenshot):

priority	customer_id	region	tenure_months	monthly_charges	support_tickets_6m	late_payments_12m
1	CUST103893	East	38	1,162	3	0
2	CUST103384	South	36	1,144	3	3
3	CUST106070	West	36	1,115	3	0
4	CUST103958	North	24	1,114	3	0
5	CUST101807	South	17	1,109	3	0
6	CUST100862	South	71	1,086	3	0
7	CUST101052	South	7	1,085	3	1
8	CUST103443	South	55	1,079	3	1
9	CUST105166	North	52	1,067	4	2
10	CUST101775	South	40	1,059	3	0

Insight: A ready-made call list: active, high-value month-to-month Fiber customers who already raised 3+ tickets.